

DBSCAN Advantages & Limitations

Advantages

1. Automatically Finds Number of Clusters

Unlike K-Means,

No need to specify K.

DBSCAN automatically discovers clusters.

2. Robust to Outliers

Noise points are automatically detected.

Example

Cluster

• • • •

x

"x" is labeled as Noise.

Very useful in

- Fraud Detection
- Anomaly Detection
- Intrusion Detection

3. Detects Arbitrary Shapes

Can cluster

- Circular
- Spiral
- Curved
- Concentric circles
- Any irregular shape

Unlike K-Means.

4. Only Two Hyperparameters

Easy to understand.

Only tune

- ϵ
- MinPts

Limitations of DBSCAN

1. Sensitive to Hyperparameters

Small change in

- ϵ
- MinPts

can completely change clustering.

Choosing these values carefully is very important.

2. Struggles with Different Densities

Suppose

Dense Cluster

Sparse Cluster

* * *

One ϵ cannot work well for both.

Performance decreases when cluster densities vary significantly.

3. No Prediction Capability

Unlike supervised learning,

DBSCAN has **no predict () method**.

If a new point arrives,

New Point ?

The algorithm cannot directly assign it to an existing cluster.

Instead, DBSCAN must be run again on the dataset.

Scikit-learn Implementation

```
from sklearn.cluster import DBSCAN

dbscan = DBSCAN(
    eps=0.5,
    min_samples=5
)

dbscan.fit(X)

labels = dbscan.labels_
```

Meaning of Labels

- 0 → Cluster 1
- 1 → Cluster 2
- 2 → Cluster 3
- -1 → Noise (Outlier)

Interview Questions

Q1. What does DBSCAN stand for?

Density-Based Spatial Clustering of Applications with Noise.

Q2. Which hyperparameters are used?

- ϵ (Epsilon)
- MinPts (Minimum Points)

Q3. What are the three types of points?

- Core Point
- Border Point
- Noise Point

Q4. Why is DBSCAN better than K-Means?

- No need to specify K
- Handles outliers
- Detects arbitrary-shaped clusters
- Density-based instead of centroid-based

Q5. What label is assigned to noise points in Scikit-learn?

-1

Quick Revision Sheet

Feature	K-Means	DBSCAN
Need number of clusters	✓ Yes	✗ No
Handles outliers	✗ Poor	✓ Excellent
Works on arbitrary shapes	✗ No	✓ Yes
Based on	Centroids	Density
Hyperparameters	K	ϵ , MinPts
Noise Detection	✗ No	✓ Yes
Sensitive to outliers	✓ Yes	✗ No
Prediction for new data	✗ No (unsupervised)	✗ No
Best For	Spherical clusters	Arbitrary-shaped clusters with noise